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Strangers in a Strange Land: A Psychobiological Study of Infant Monkeys Before and After Separation from Real or Inanimate Mothers

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KRAEMER, GARY W.; EBERT, MICHAEL H.; SCHMIDT, DENNIS E.; and MCKINNEY, WILLIAM T. *Strangers in a Strange Land: A Psychobiological Study of Infant Monkeys Before and After Separation from Real or Inanimate Mothers.* CHILD DEVELOPMENT, 1991, **62**, 548–566. Some rhesus monkey (*Macaca mulatta*) infants have a "despair" or depression-like response to mother-infant separation, while others do not. The presumed interrelation between early rearing conditions and the neurobiological status of the infant that might lead to increased risk for despair is not understood. In this study, the characteristics of the "mother" were controlled by rearing infant rhesus monkeys with their biological mothers, or with inanimate mothers. Behavioral data were collected before and after separation at 6–7 months of age. The neurobiological status of the infants was evaluated by measuring the concentration of norepinephrine, its major metabolite, and the metabolites of dopamine and serotonin in cerebrospinal fluid. The results suggest that despair is not simply a behavioral response to separation. Instead, despair may reflect the inability to cope with the separation environment. Coping with the separation environment appears to depend on neurobiological and behavioral characteristics of the infant that are related to, if not determined by, characteristics of the mother.

The behavioral response to mother-infant separation in human and nonhuman primates has often been characterized as occurring in two sequential phases, "protest" and "despair." Protest is manifested by increased activity, agitation, and vocalization. Despair is characterized by inactivity, withdrawal, and often failure to eat (Bowlby, 1960; Spitz, 1946). In despairing rhesus monkeys, there is also an increase in self-directed behaviors such as self-mouthing (finger and toe sucking) and self-clasping, and a characteristic adoption of a fetal-like self-enclosed body posture (huddling) (Harlow, Harlow, & Suomi, 1971; Kraemer & McKinney, 1979; Mineka & Suomi, 1978). However, separations may be critical events for some individuals and not others (Rutter, 1979b). Only a small proportion of human and rhesus monkey infants that are separated from their mothers during the stage of early dependency subsequently have a despair response (Kraemer & McKinney, 1979; Lewis, McKinney, Young, & Kraemer, 1975; Spitz, 1946). The question this study addresses is why some rhesus infants have a despair response to mother-infant separation and others do not.

The answer to this question has both clinical and theoretical significance because disrupted mother-infant attachment and/or premature separation have been cited as factors that may promote vulnerability to stressors and the development of psychopathology later in life (Akiskal & McKinney, 1975;

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Bowlby, 1982; Kaufman, 1977; Paykel & Hollyman, 1984; Reite & Capatanio, 1985). While separation is often the precipitating cause of despair, the low proportion of infants afflicted suggests that there are predisposing or underlying factors that must also be present. Since human infants do experience unavoidable separations, prevention or reversal of despair and amelioration of its probable long-term consequences depend in large measure on identifying these contributing factors so that their impact may be reduced or eliminated.

Risk Factors

Early rearing experience.—The response to separation often appears to be related to characteristics of the mother-infant relationship. Bowlby (1969, pp. 333–334) and Ainsworth (Ainsworth, Blehar, Waters, & Wall, 1978) suggested that a competent mother can be thought of as a "secure base" from which the infant may sortie to explore the world and then return to in the face of danger or physiological need. The infant forms what Bowlby (1969, p. 80) referred to as a "working model" of the mother; an internal representation of what to depend on in the present and what to expect in the future, framed in relation to needs of the infant that the mother can fulfill. As a simplification, infants characterized as being securely "attached" to their mothers cope with separations quite well. Insecurely or anxiously attached infants do not cope well with separations (Ainsworth et al., 1978; Bretherton, 1985). In laboratory-reared monkeys, there is an analogous range of variation in the response to separation, and the nature of the response appears to be related in part to characteristics of the mother. Infants reared by rejecting or inattentive monkey mothers have higher levels of disturbance behavior than infants reared by attentive and compassionate monkey mothers (Arling & Harlow, 1967; Harlow et al., 1971; Hinde, 1977; Rosenblum & Paully, 1984).

In order to identify the characteristics of the mother that are most relevant to mother-infant attachment and infant well-being, monkeys have also been reared with terry cloth covered pedestals ("surrogate mothers"), with surrogates that move or rock, with dogs, or with surrogates equipped to repel the infant in various ways. The surrogate must have contact comfort characteristics before the infant will exhibit attachment behaviors (Harlow, 1958). Beyond this, infants reared with moving surrogates or dogs are able to cope with separations and/or novel environments better than infants reared with immobile surrogates (Anderson, Kenney, & Mason, 1977; Mason & Berkson, 1975; Mason & Capitanio, 1988). Infants reared with rejecting surrogates are most prone to despair (Harlow et al., 1971; Harlow, Plubell, & Baysinger, 1973). Thus, the response to separation is modified by the nature of the surrogate; salient variables are whether the surrogate stimulates and/or is responsive to the infant and does not physically reject it.

Separation environment.—Separated infants that have access to alternative attachment objects are less prone to despair than infants confronted with environments lacking alternative attachment objects (Chappell & Meier, 1975; Kaufman & Stynes, 1978; Mason & Capitanio, 1988; Mason & Kenney, 1974; Reite, Kaemingk, & Boccia, 1989).

Neurobiological development.—Altered brain norepinephrine (NE) system function may be associated with risk for despair. Brain NE system function appears to be associated with reinforcement, "tuning" of responses to stressors, circadian rhythms, and neuronal plasticity (Foote & Morrison, 1987a, 1987b; Glavin, 1985; Iversen, 1984; Kraemer, 1988; Oades, 1985; Svensson, 1987; Weiss & Simson, 1986). In general, compromised brain NE system function has been associated with reduced reward, failure to attend to significant stimuli, failure to avoid stressors, altered sleep cycles, and reduced brain organizational plasticity (Kraemer, 1986). The first four of the characteristics associated with reduced NE system function are also characteristic of despair, and it has been hypothesized that reduced organizational plasticity may be the core neurobiological substrate of despair (Kraemer, 1986).

The brain NE system continues to develop postnatally in primates (Foote & Morrison, 1987a; Goldman-Rakic & Brown, 1982), and the characteristics of the social environment appear to be a factor in this development (Kraemer, Ebert, Schmidt, & McKinney, 1989). The concentration of NE in cerebrospinal fluid (CSF) is thought to vary with the activity of the brain NE system, and the collective results of a series of studies indicate that: pharmacological agents that reduce CSF NE augment separation-induced despair, especially in maternally deprived monkeys; juvenile monkeys with the lowest levels of CSF NE during separation are most likely to exhibit despair; and, maternal deprivation is associ-

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ated with low levels of CSF NE in infancy (Kraemer, Ebert, Lake, & McKinney, 1984b; Kraemer & McKinney, 1979; Kraemer et al., 1989). Thus, there appears to be a correspondence between reduced CSF NE and despair, and different rearing conditions may be associated with either the production or expression of the characteristic of having relatively low CSF NE (Kraemer, 1984a, 1984b; Kraemer, 1988).

Environment and Neurobiology

Two components of risk for despair that can exist before a separation might occur in infancy are: (1) the characteristics of the mother and the mother-infant relationship, and (2) the neurobiological status of the infant. However, it is not known to what degree neurobiological characteristics of the infant, and, to be specific, the level of CSF NE, depend on characteristics of the mother-infant relationship, and to what degree the neurobiological characteristics of the infant might be largely impervious to differences in maternal care and could form an independent risk or protective factor.

In previously published accounts of a longitudinal study we concluded that maternal rearing produced persistent differences in the level of CSF NE in rhesus infants, and that subsequent changes in CSF NE were associated with both the formation and disruption of social attachments (Kraemer et al., 1989). These conclusions were based on comparison of mother-reared infants and infants reared without an attachment object of any sort (isolated/mother-deprived). Mother-deprived infants have low levels of CSF NE by comparison to mother-reared infants, and this does not appear to be related to nutrition or other variations in physiological development attributable to different rearing conditions (Kraemer et al., 1989). This suggests that low CSF NE in mother-deprived infants might be attributable to lack of an attachment object and associated attachment behaviors, while the increased level of CSF NE in mother-reared rhesus infants could be related to some aspect of mother-infant attachment.

Objectives of the Present Study

The first objective was to compare CSF NE and measures of activity in other brain neurotransmitter systems in rhesus infants reared with surrogate or monkey mothers. Since infants reared in either condition exhibit comparable attachment behavior, this would allow us to determine whether attachment behavior on the part of the infant produces increases in CSF NE regardless of the characteristics of the "mother." Comparison of neurobiological characteristics of infants reared by real versus inanimate mothers also provides some indication of the degree to which real mothers might produce neurobiological changes in the infant.

The second objective was to determine whether behavioral and/or neurobiological characteristics of the rhesus monkey infant prior to separation are related to its behavioral and neurobiological response to separation, and then whether there was any relation between the behavioral and neurobiological response to separation that might indicate the two were interdependent. This would allow us to estimate the degree to which variation in the neurobiological characteristics of the infant might contribute to behavioral variation in the response to separation.

Method

Subjects

Twenty-six rhesus monkeys (*Macaca mulatta*) were used in this study.

Rearing Conditions

Mother rearing.—Twelve male rhesus infants were reared with their biological monkey mothers until they were 6–7 months of age. The mothers and infants were provided with Purina Monkey Chow once per day at a standard ration for nursing mothers (5%–6% of maternal body weight). Water was available ad libitum. Hereafter, these infants are referred to as "mother-reared."

Surrogate rearing.—Eight (four male, four female) rhesus infants were separated from their biological mothers within 3 days of birth and housed in individual cages in a nursery for the next 30 days. They could see, hear, and smell other infants but could not touch them. The infants were fed with infant formula (Similac) four times per day. All eight infants were provided with a terry cloth covered surrogate mother, and hereafter are referred to as "surrogate-reared."

One objective was to provide surrogates that differed in their "responsiveness" to the infant. Therefore, four infants (two males, two females) were housed with a spring-loaded surrogate that would rock with movements of the infant. These are referred to as "rocking surrogates." The other four infants (two males, two females) were housed with surrogates that were rigidly mounted to their base. These are referred to as "stationary surrogates."

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At 30 days of age, the infants and their surrogates were transferred to standard individual housing cages. Beginning at 1 month of age, the infants were allowed to interact with each other for 1 hour per day, 3 days per week. Purina Monkey Chow was provided once per day at a ration of 4%–5% of body weight. In months 2–3, 150 ml of infant formula was provided in the morning (0800) and evening (1700). In months 4–6, infant formula was provided only in the evening. Dietary supplement of infant formula was discontinued in month 6.

Mother deprivation.—Six male rhesus monkeys were reared without attachment objects. They were separated from their mothers within 2 days of birth and housed in individual wire mesh cages in a nursery for the next 30 days. They could see, hear, and smell other infants but could not touch them. The dietary regimen for these infants was identical to that of the surrogate-reared infants. Hereafter, these monkeys are referred to as "mother-deprived."

All environments were regulated on a 0600–1800 light, 1801–0559 dark cycle.

Dependent Measures

Data were collected on the mother-reared and mother-deprived groups beginning at 1 month of age (Kraemer et al., 1989). Since the mother-deprived infants did not have any attachment object, they were not part of the separation study; their neurochemical data are presented to provide a comparison group for the mother- and surrogate-reared infants.

The surrogate-reared infants, on the other hand, were part of another project examining the interaction between rearing conditions and behavioral measures thought to be related to individual differences in "temperament" (Schneider, 1987). These monkeys were not available for study prior to the present experiment, so the data for the surrogate-reared infants are limited to several weeks before and after separation from their surrogates.

Neurobiological measures—cerebrospinal fluid (CSF) collection.—The concentration of NE, 3-methoxy-4-hydroxyphenylglycol (MHPG), 5-hydroxyindoleacetic acid (5HIAA), homovanillic acid (HVA), and 3,4 dihydroxy-phenylacetic acid (DOPAC) in CSF were measured before and after mother-infant separation. MHPG and 5HIAA are the major metabolites of NE and serotonin, respectively. HVA and DOPAC are the major and minor metabolites of dopamine, respectively.

CSF was always collected at the same time of day (1330–1400). Approximately 1.0 ml (0.8–1.2 ml) was withdrawn from the cisterna magna using a 25-gauge, 5/8-inch hypodermic needle and syringe. Ketamine anesthesia (10 mg/kg) was used to limit movement of the monkeys during cisternal puncture. The mother-reared infants were removed from their cage with their mother, and both the mother and infant were anesthetized. This eliminated the usual trauma associated with separating mothers and infants for sample collection. The surrogate-reared and mother-deprived infants were captured directly out of their cage, injected with ketamine, and returned to their cage until the anesthetic effect was evident (4–6 min). The monkeys returned to their usual patterns of social behavior within hours and without signs of discomfort or impairment. Ketamine does not affect NE or the acid metabolite measures within 15 min of injection (Bacopoulos, Redmond, & Roth, 1979; Chernow et al., 1982).

Assay methods (NE, MHPG, DOPAC, 5HIAA and HVA).—The CSF samples were stored at −70 C, and later analyzed by HPLC methods as described by Kraemer et al. (1989). The minimum sensitivity was approximately 50 fmoles for MHPG, DOPAC, and NE, and 100 fmoles for HVA and 5HIAA. The coefficient of variation for repeated between-day analyses ($N = 10$) of a pooled test sample of primate CSF was 2.3%–4.5%. The levels of free dopamine and serotonin in CSF were below the minimum sensitivity of the assay.

Behavioral measures.—The behavioral scoring system used 45 operationally defined behavioral scoring categories. The definitions of behavioral categories of importance in the present study are presented in Table 1. The categories themselves are either the same as, or expansions on, behavioral scoring systems in place at the Harlow Primate Laboratory since the mid-1950s (see Harlow, 1958; Meyer, Novak, Bowman, & Harlow, 1975). The combination of behaviors that are collectively taken to be indicative of "protest" and "despair" are also listed in Table 1.

The behavioral data collection apparatus was an APPLE II microcomputer with the features of the encoding process, randomization of scoring order, recording of

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TABLE 1

DEFINITIONS OF BEHAVIORAL SCORING CATEGORIES AND CLUSTERS INDICATIVE OF PROTEST AND DESPAIR

Individual behaviors:	
Environmental exploration	Any manual or oral exploration or manipulation of the physical surroundings
Inactive	Maintenance of a stable, stationary position with no simultaneous scorable behavior, with the exception of climbing and vocalization
Locomotion	Ambulation of one or more full steps at any speed. Includes jumping
Motor stereotypy	Repetitive, patterned, and rhythmic movement paired with locomotion
Self-directed behavior	Manual or pedal clutching or grasping of self, or oral nonaggressive contact with any part of own body, that is not part of ongoing behavior such as self-grooming or self-aggression
Huddling	Maintenance of a self-enclosed fetal-like body posture with head at or lower than shoulder level
Rocking	Sustained nonlocomotor back-and-forth motion of whole body. Scored with or without contact with attachment object
Maternal contact	Manual, pedal, or whole body contact with attachment object
Vocalization	Any audible sound produced by the monkey fitting into the general classes of cooing, barking, or screeching (in this study, virtually all vocalizations were coos or screeches)
Behavior clusters:	
Protest	Vocalization
	Locomotion
	Motor stereotypy
Despair	Huddling
	Rocking
	Self-directed nonaggressive behavior

time of day, data evaluation, and storage controlled by software.

In this study, one observer scored the infants while they were with their attachment object, and another observer scored the infants while they were separated. Technically, then, the observer of the separated infants did not know whether they were reared with monkey mothers, or stationary or rocking surrogates. In actuality, the behavior patterns of infants reared in different conditions are so different that experienced observers rapidly discriminate them on the basis of rearing condition. It is not claimed that this source of bias was controlled, nor that it could be controlled.

Observer reliability is also somewhat of a moot point. Behaviors requiring duplication of judgment, and for which reliability might be an issue, were scored by different observers. Those behaviors that could occur in all conditions, such as locomotion, huddling, or inactivity, are unambiguous. The trained observers used in the present study, each having at least 3 years of experience, never have had frequency and/or duration reliability coefficients less than 0.90.

Statistical methods.—Neurochemical and behavioral data were evaluated using mixed-design ANOVA with rearing groups as a between-subjects variable. The results of these analyses will be presented in the figures and tables, or in the text where main effects, interactions, or comparisons are too cumbersome to diagram on the figures and tables. Tukey *B* tests were used for post hoc point contrasts. Relations among measures were assessed using Pearson product correlations and multiple regression analysis where appropriate (Zar, 1984).

Experimental Protocol

Separation environment.—Separated infants were housed individually in 1 m × 1 m × 1 m stainless steel sheet-metal cages

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with a wire mesh front panel. Four cages were ordered in a row and separated by Plexiglas panels. The infants could see, hear, and smell other infants, but could not touch them. The monkeys were housed in a room dedicated to this study; there were no other monkeys in the room, and there was no exposure to humans other than observers and animal care staff during separation. The room temperature and lighting control were the same in all conditions, that is, there were no changes in these environmental factors associated with separation.

Mother-infant separation.—There is a marked increase in maternal rejection behavior when the rhesus infant reaches 6–7 months of age (Harlow, Harlow, & Hansen, 1963). Therefore, the mother-infant relationship, and thus what the infant is being separated from, may differ considerably with age. Nevertheless, as part of the longitudinal study, it was necessary to leave the infants with the mothers up to this point to insure that they experienced as much of the full spectrum of mother-infant interaction as possible.

The mother-reared infants (N = 12) were born over a 2.5-month range. It was not possible to separate all of the infants at the same age and then observe them for 2–4 weeks because the separation environment would only accommodate four infants at a time. Therefore, the following protocol was followed. The infants were separated from their mothers in groups of four infants each. The first group was separated from their mothers at 6 months of age. Except for behavioral observation, the infants were left undisturbed for 4 weeks. Behavioral data were collected twice per day between 1200–1230 and 1500–1530. Each infant was observed for 5 min per session. There was no evidence of a despair response. At 4 weeks postseparation a CSF sample was collected, and this did not alter the ongoing behavior of the infants over the next week. We concluded that collection of CSF prior to separation was not likely to potentiate a despair response, and that it would be valuable to collect postseparation CSF at a time closer to the separation event. This assessment took more than 1 month to complete, and some infants remaining with their mothers were nearing 7 months of age.

Therefore, the age of separation was balanced so that 6 out of 12 mother-reared infants were separated at 6 months of age and 6 were separated at 7 months of age. Behavioral data were collected as before, and CSF samples were collected 2 and 4 weeks after separation.

Only one infant had a despair response, and this emerged 2 days after separation. Therefore, 2 weeks of undisturbed behavioral data collection and the mean values recorded were used to document the response to separation.

Surrogate-infant separation.—In this study, as in many other studies of mother-infant separation, the infant was separated from the mother and placed in a new environment. In this case it is likely that the response observed is not entirely due to separation. The infant is also responding to the new environment (Hinde, 1977). However, it is not possible to determine how neurobiologically responsive mother-reared infants are to new environments prior to separation. Environmental changes alter the behavior of the mother. Thus, modest changes in the infant's behavior occur if the mother and infant are transferred to a new environment, but these appear to be attributable to changes in the mother rather than the environment (McGinnis, 1978). The surrogate-reared infants presented the possibility of documenting the infant's response to environmental transitions because the "behavior" of the surrogates is constant in the face of environmental transition. Therefore, this aspect of the study documents the behavioral and neurobiological response of the infant to environmental transitions with a mother that does not respond to the transition, and then a separation of the infant into the same environment.

It must be noted that there were four females in the surrogate-reared condition, so sex is confounded with rearing condition in comparison with the other all-male groups. There are no reports of significant sex differences in the response to mother-infant separation in rhesus monkeys. However, there are reports that males are more "distressed" following brief separations at about 4–5 months of age (Hinde & Spencer-Booth, 1971). We also have a considerable body of unpublished data indicating that males are more likely to have a despair-like response to separation, but the magnitude of this difference is very low and only becomes statistically significant in considering populations of 60/60 male/female infants or more. Therefore, this study may be biased against finding a despair response following separation in the surrogate-reared group.

The surrogate-reared infants were treated as two groups of four monkeys each,

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with two males and two females in each group. Within the variable of gender, one monkey was reared on a rocking surrogate and one on a stationary surrogate.

The surrogate-infant separation study was conducted in three phases. Initially, all of the surrogate-reared infants were individually housed in wire-mesh cages in one room. The infants could see, hear, and smell other infants and could touch those in adjacent cages through the mesh. As noted before, they had 3 hours per week of social interaction with each other.

In phase 1, when the mean age was 5 months, social interaction was canceled. In phase 2, when the mean age was 6 months, each group was transferred from their normal housing room to the separation environment with their surrogate for 7 days (Monday through Sunday). As with the mother-reared infants, behavioral data were collected twice per day between 1200–1230 and 1500–1530. Each infant was observed for 5 min per session. CSF samples were collected on the Friday preceding housing change and on the last day of housing change, after which the infants were returned to their normal environment. In phase 3, when all infants were approximately 7 months of age, each group was transferred from their normal housing room to the separation environment without their surrogate. CSF samples were collected before and 2 weeks after separation as they were for the mother-reared infants. Behavioral data were collected as described before.

Results

Neurobiology

There were no significant between-group differences or effects of separation on CSF HVA, DOPAC, or 5HIAA. Figure 1 shows the concentration of CSF NE in mother-reared and surrogate-reared rhesus infants before and after separation or housing transition. Also shown are the CSF NE values for mother-deprived infants.

Surrogate-reared and mother-deprived infants had lower CSF NE than the mother-reared infants, $F(1,18) = 4.20$, $p < .05$; $F(1,12) = 4.83$, $p < .05$, respectively. There was no difference in the baseline CSF NE values between the surrogate-reared and mother-deprived infants. Therefore, provision of a surrogate and 3 hours per week of

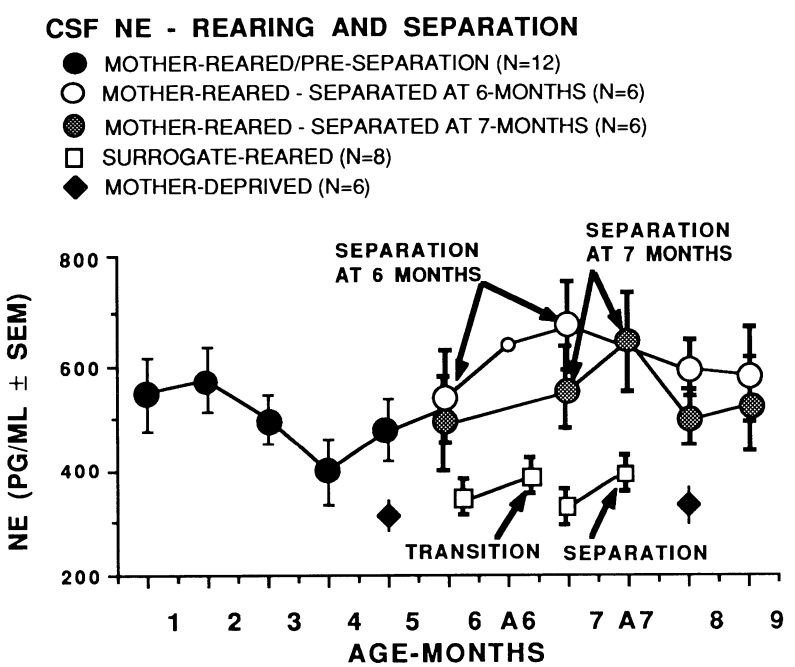


FIG. 1.—Concentration of NE of different rearing groups before and after housing transition (surrogate-reared infants), separation from attachment object (mother- and surrogate-reared infants), or in infants with no attachment object (mother-deprived infants). Notations on the abscissa (A6 and A7) denote collection of CSF samples 2 weeks (A)fter separation at 6 or 7 months of age.

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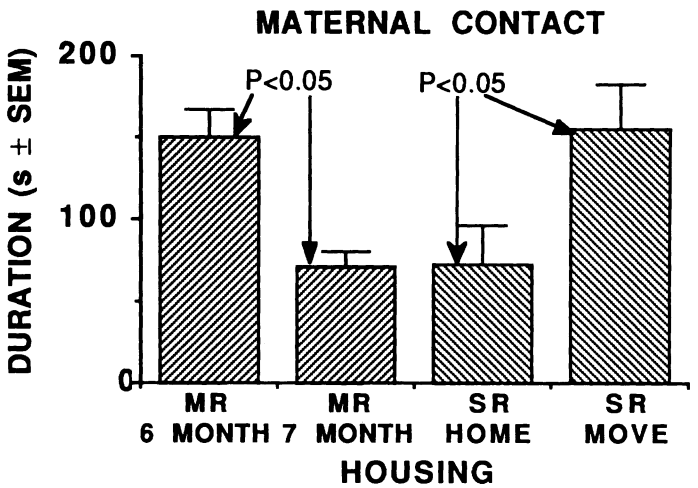


FIG. 2.—Duration of physical contact with mother in mother-reared (MR) infants prior to separation at 6 (MR-6) or 7 (MR-7) months of age, and duration of physical contact with the surrogate in surrogate-reared (SR) infants prior to surrogate infant separation (SR-HOME) or during housing transition (SR-MOVED).

playtime with peers did not elevate CSF NE levels over and above those observed in the mother-deprived infants.

Separation from the mother or surrogate produced an increase in CSF NE, $F(1,11) = 6.00$, $p < .03$; $F(1,7) = 6.05$, $p < .03$. This increase persists indefinitely if the infants are not returned to a social environment (Kraemer et al., 1989).

In the surrogate-reared group, housing transition also produced an increase in CSF NE, and this was not significantly different from that produced by surrogate-infant separation.

In the mother-reared infants that were not separated from their mothers until 7 months of age, CSF NE increased at 6 months of age, $F(1,5) = 6.72$, $p < .05$, and then increased even further following mother-infant separation. The infants also spent less time in contact with their mothers in the seventh postnatal month that they had over the previous 6 months (Fig. 2).

The surrogate-reared infants had lower CSF MHPG than both the mother-reared infants and the mother-deprived infants, $F(1,18) = 10.65$, $p < .004$, $F(1,12) = 9.40$, $p < .01$, respectively, data not shown. Separation did not have a lasting effect on CSF MHPG in the infants separated at 6 months of age, but produced a persistent increase in CSF MHPG in infants separated at 7 months of age, $F(1,5) = 24.2$, $p < .001$. Therefore, the effects of rearing condition and separation on CSF MHPG were less consistent and generalizable than the effects on CSF NE.

There was no difference in the neurochemical response to separation among infants reared on rocking as opposed to stationary surrogates, nor were there any main or interactive effects of sex in the surrogate-reared group.

Behavior

Figure 3 (*a*, *b*, *c*) shows the data for behaviors indicative of protest and despair, and the data for two other behaviors (environmental explore and inactivity) that collectively account for most of the behavior of individually housed monkeys.

Surrogate-infant environmental transition.—Moving surrogate-reared infants with their surrogates produced increases in vocalization, self-directed nonaggressive behavior, huddling, and rocking. Housing transition also produced an increase in surrogate contact (Fig. 2). So, behavioral disturbance often occurred while the infant was in contact with the surrogate. There were no changes in the duration of locomotion, environmental explore, or inactivity.

Surrogate-infant separation.—Inactivity and vocalization increased during surrogate-infant separation. The mean durations of all other disturbance behaviors were comparable to those observed during environmental transition. There was no significant main or interactive effect of gender of the infants in any condition.

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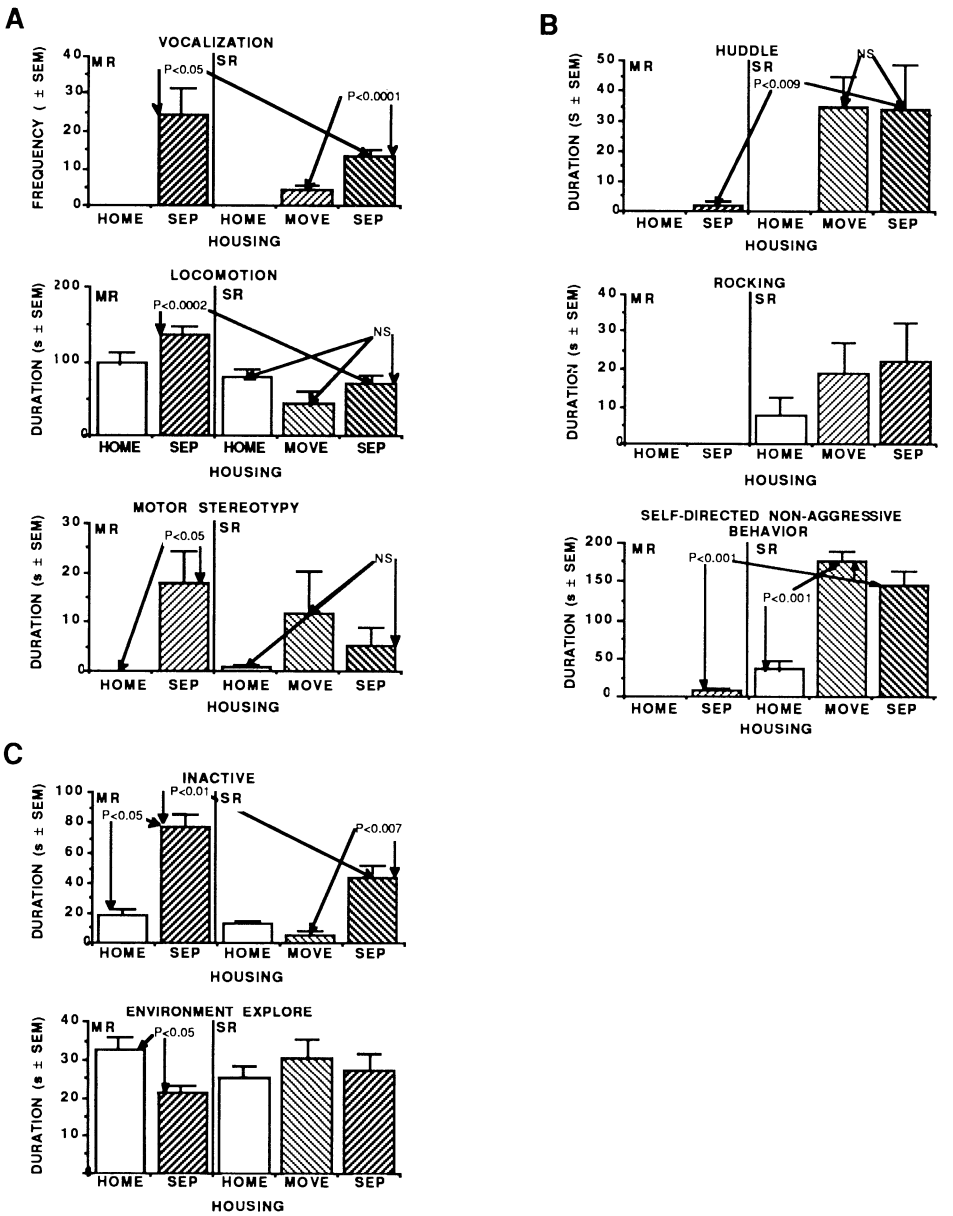


FIG. 3.—Behavior durations before (HOME) and after separation (SEP) or housing transition (MOVE) in mother-reared (MR) and surrogate-reared (SR) infants. Definitions of behaviors are presented in Table 1. *A*, Behaviors indicative of protest; *B*, Behaviors indicative of despair; *C*, Behaviors not associated with protest or despair.

Rocking versus stationary surrogates.—Figure 4 (*a, b, c*) shows the data for the behavioral response to environmental transition or separation of infants reared on a rocking or stationary surrogate. There were differences in the duration of environmental exploration, vocalization, and rocking attributable to the characteristics of the surrogate alone. Infants reared on rocking surrogates explored more and rocked less whether they were separated or not. Infants reared on rocking surrogates were activated by separation, and they markedly increased their distress vocalizations, that is, their response was more protest-like than infants reared on stationary surrogates.

Mother-infant separation.—Mother-infant separation produced increases in locomotion, stereotypy, inactivity, vocalizations,

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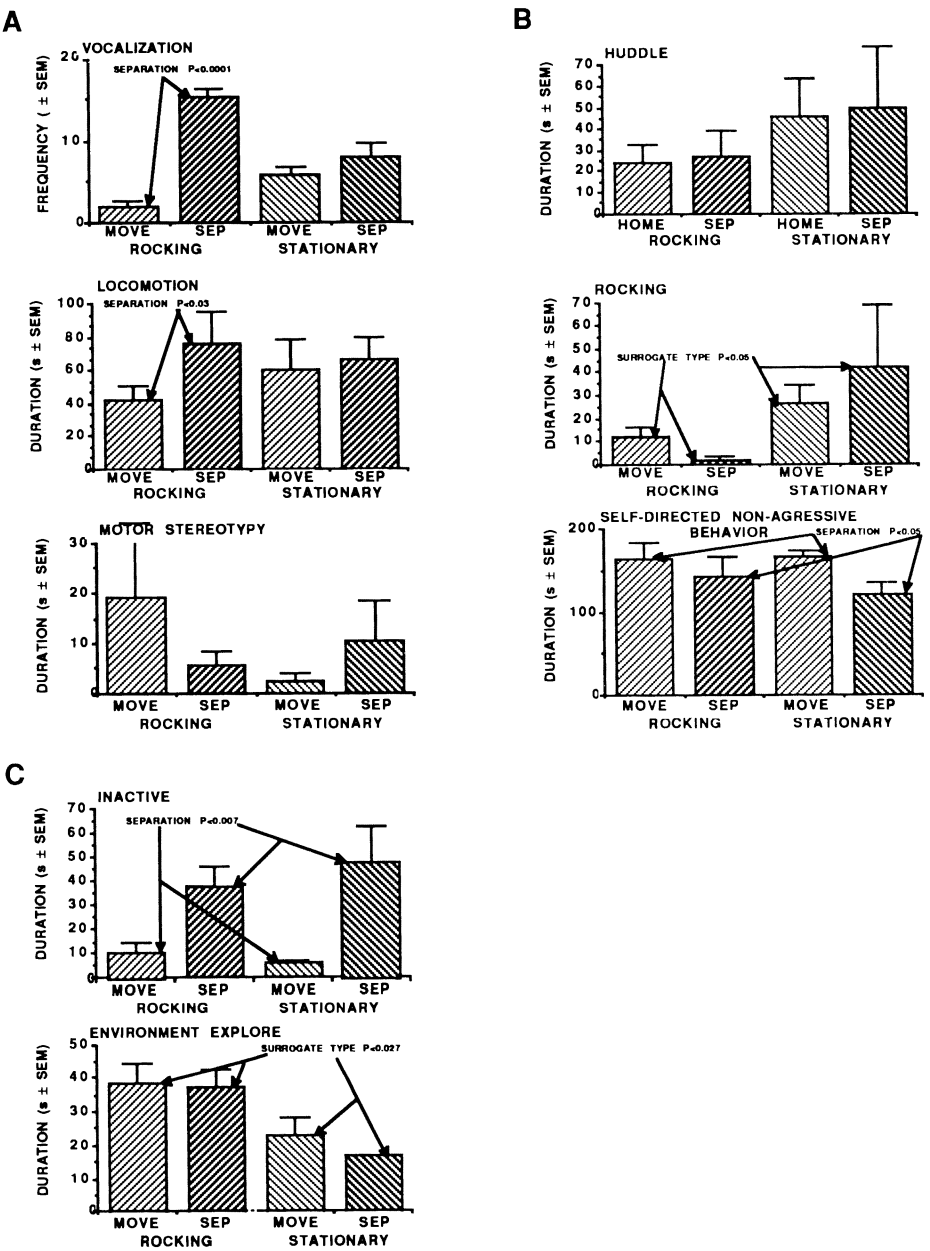


FIG. 4.—Behaviors altered by rearing with rocking (ROCK) versus stationary (STAT) surrogates during housing transition (MOVE) or following surrogate-infant separation (SEP). Operational definitions of the behaviors are presented in Table 1. Statistical comparisons are broken down to indicate changes in behavior that were attributable to SEPARATION and then whether SURROGATE TYPE altered the response to separation. *A*, Behaviors indicative of protest; *B*, Behaviors indicative of despair; *C*, Behaviors not associated with protest or despair.

and reductions in environmental exploration, that is, there was a "protest" response to mother-infant separation. One monkey had a despair response peaking at about 72 hours after separation, and this lasted over the next several days. However, in the mother-reared group taken as a whole there were no significant increases in self-directed nonaggressive behaviors, huddling, or rocking.

Comparisons between separated mother-reared and separated surrogate-reared infants.—The data presented in Figure 3 in-

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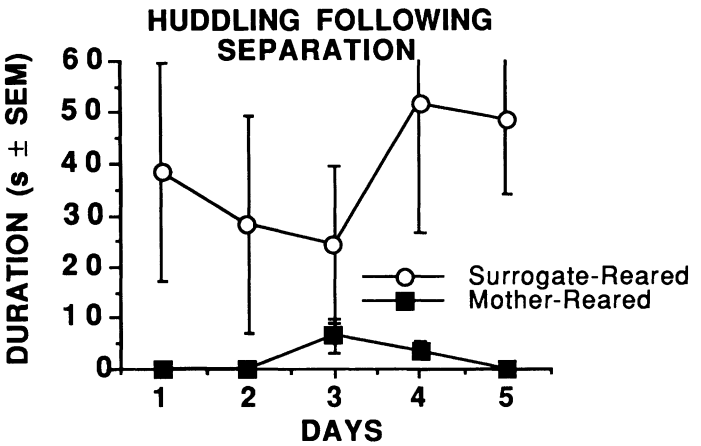


FIG. 5.—Duration of huddling over the first 5 days of separation in mother-reared and surrogate-reared infants. Huddling is defined in Table 1.

dicate that mother-reared infants divided their time into either activity (i.e., locomotion and motor-stereotypy, or inactivity). In contrast, the surrogate-reared infants spent a significantly larger proportion of time engaged in self-directed behavior, huddling, and rocking.

Figure 5 shows the data for huddling in the surrogate-reared and mother-reared infants for the first week of separation (5 days of observation). Huddling in the surrogate-reared infants began on day 1. There was no transition from a protest stage with low levels of huddling to a despair stage characterized by huddling.

Behavioral/Neurochemical Relations

Correlations.—In general, there were no significant relations between CSF HVA, DOPAC, and 5HIAA measures and behavior, with the exception that CSF HVA and 5HIAA were positively correlated with locomotor activity while the infants were separated.

One objective of correlation analysis was to determine whether there were relations between the behavioral and neurobiological characteristics of individuals prior to separation that then predicted the nature or magnitude of the response to separation.

Overall, the surrogate-reared infants had the lowest levels of CSF NE and the highest levels of despair behavior. Thus, correlations using all mother- and surrogate-reared subjects yield statistically significant correlations between CSF NE and despair behaviors between −0.80 and −0.90. This relation is attributable to rearing-group differences. The next question was whether there were behavioral/neurochemical relations that existed independently of rearing conditions and could be attributable to genetic or perinatal factors.

Within-rearing-groups correlations.—Behaviors associated with the response to separation were correlated predominantly with changes in CSF NE in relation to baseline values. There were no significant behavior/MHPG correlations. Huddling is the behavior often taken to be most characteristic of a despair response, so the relation of this behavior to CSF NE will be considered first. Table 2 shows the relations between huddling and CSF NE in the mother- and surrogate-reared groups.

Within the surrogate- and mother-reared groups, the correlation between CSF NE concentration prior to separation and the duration of huddling following separation was negative, but neither correlation was statistically significant. Since separation produces increases in CSF NE in most infants, the next question was whether the change in CSF NE associated with separation was correlated with the baseline value (postseparation pg/ml − preseparation pg/ml = change score). In both the mother- and surrogate-reared groups, the change in CSF NE was negatively correlated with the baseline value. Thus, infants with the lowest baseline values had the largest separation-induced increases in CSF NE. This could be attributed to a "ceiling" effect. However, data from previous (Kraemer, Ebert, Lake, & McKinney, 1984a) and ongoing studies indicate that CSF NE levels can be driven to at least double the levels observed in the

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TABLE 2

CORRELATIONS BETWEEN PRESEPARATION CEREBROSPINAL FLUID NE, CHANGE IN NE, AND HUDDLING IN MOTHER- AND SURROGATE-REARED GROUPS

NE and Huddling	Mother Reared	Surrogate Reared
Preseparation NE vs. huddling	−.49	−.67
Preseparation NE vs. change in NE	−.79*	−.83*
Change in NE vs. huddling	.94*	.67

* = $p < .05$. There were no significant between-group differences in correlation coefficients.

present study by other stressors or by treatment with d-amphetamine. So, the negative correlation of increase in CSF NE with baseline level may not be attributable to biological limits of the system.

Increases in CSF NE and duration of huddling were positively correlated. For the mother-reared infants, this correlation is deceptive because one monkey huddled much more than the others and had the largest increase in CSF NE. Therefore, the next question was whether the change in CSF NE was correlated with the behaviors that collectively reflect the magnitude of protest or despair in each rearing group.

Multiple linear regression analysis was used to determine whether the increase in CSF NE was correlated with the cluster of behaviors indicative of protest (vocalization, locomotion, motor-stereotypy) or despair (huddling, rocking, and self-directed nonaggressive behavior). However, a problem with assessing individual differences in the "magnitude" of despair is that some individuals may huddle, while another may huddle, rock, and self-clasp and self-mouth in some combination, or alternate between behaviors. The latter individual may not be conspicuous with regard to the duration of any particular behavior, but perceptually is the most disturbed. Therefore, the correlation between the change in CSF NE and the additive sum of the durations of the behaviors associated with protest or despair was also determined. The multiple and single correlations are presented in Table 3. A general conclusion is that changes in CSF NE were correlated with the behavioral clusters associated with both protest and despair in the mother-reared infants, and with despair in the surrogate-reared infants.

Individual differences.—Given the one mother-reared infant that did have a despair response, it is worthwhile to describe this individual in relation to the other mother-reared infants. Infants with an abbreviated protest response to separation and the infant displaying a despair response to separation had low levels of mother-infant contact prior

TABLE 3

CORRELATIONS BETWEEN BEHAVIORS INDICATIVE OF PROTEST AND DESPAIR AND CHANGE IN LEVEL OF NE IN CEREBROSPINAL FLUID

Behavior	Mother Reared		Surrogate Reared	
	r_m	r_{add}	r_m	r_{add}
Despair:				
Huddle, self-directed behavior, rocking	.95*	.96*	.67	.84*
Protest:				
Stereolocomotion, locomotion, vocalization[a]	.85*	.77*	.69	.53

NOTE.—Behaviors associated with protest and despair are defined in Table 1. Multiple correlation coefficient = r_m. Correlation of additive sum of behavior durations with change in NE = r_{add}.

* = $p < .05$. There were no significant between-group differences in correlation coefficients.

[a] Frequency not added into durations.

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to separation. Following separation, infants evincing minimal behavioral responses to separation (protest) also had minimal CSF NE changes. The infant that exhibited a despair response had the lowest level of CSF NE prior to separation and a large increase in CSF NE following separation. This infant was bitten several times by its mother during months 4–6 of rearing.

Discussion

The major findings of the present study in relation to the objectives, and the results of previous studies, are:

1. Separation produced increases in CSF NE in both mother- and surrogate-reared infants. However, CSF NE increased at 6 months of age in infants that remained with their mothers and when surrogate-reared infants were transferred to a new housing environment with their surrogates. Therefore, separation per se is probably not the major cause of increases in CSF NE. In the surrogate-reared infants, increases in CSF NE can be produced by environmental changes alone.

In the mother-reared infants, it is not clear how much of this neurobiological response is due to separation or "loss" and how much is due to environmental transition. However, later, when these infants were combined into peer groups as part of a longitudinal study, there was either a reduction, or no change, in CSF NE associated with this housing transition, even though this social environmental change demands and/or precipitates major alterations in activity and behavior (Kraemer et al., 1989). Therefore, environmental changes per se, even quite dramatic ones, do not necessarily produce increased CSF NE in mother-reared infants. This leads to the speculation that increases in CSF NE are associated with social environmental changes that the infant is not prepared for either by genetic endowment or prior rearing experience. Maternal rejection or separation into an environment lacking peer-aged monkeys appear to cause this response.

2. Mother-reared infants exhibited a protest response and surrogate-reared infants exhibited a despair-like response to separation. CSF NE increased in both groups, so the difference in the behavioral response is not associated with differences in this neurobiological response to separation.

3. Preexisting neurobiological characteristics of the infant are related to the nature of the behavioral response to separation. Low CSF NE was associated with increased risk for a despair response. A major determinant of CSF NE level in laboratory-reared rhesus infants appears to be the characteristics of the mother. Provision of a surrogate and access to peers does not produce CSF NE levels comparable to those observed in mother-reared infants. However, modest alterations in the characteristics of the surrogate, such as whether it rocks or is stationary, can alter the behavioral response to environmental transitions or separation.

4. The component of neurobiological risk most likely to be attributable to genetic or perinatal factors appears to be expressed in the lability or stability of behavior and CSF NE levels. Regardless of rearing condition, individuals with the largest change in CSF NE also had the largest behavioral response to separation, whether this was protest or despair.

Behavioral Attachment Systems

Ethological and also commonsensical explanations of variation in the response to separation suggest that the nature of the separation response is related to the survival role of the mother-infant relationship and "attachment" (Ainsworth et al., 1978; Bowlby, 1973). One implication of this view is that the strength of attachment should be related to the survival value of the attachment object to the infant (Harlow et al., 1963; Kaufman & Rosenblum, 1967; Rutter, 1972, pp. 13–28). Another implication is that the response to separation should be proportional to the strength of the attachment. While the strength of the underlying attachment bond may not always be revealed by "attachment behavior," collectively referring to seeking proximity, and physical and visual contact, it may be proportional to the nature and magnitude of the behavioral response if separation is threatened or sustained (Bowlby, 1969). Thus, protest, grief, and despair are often viewed as variations of a response to loss that differ in severity and time of onset, and also behaviors that reveal something of the nature and strength of attachment. In particular, despair is often likened to a state in which the infant behaves as though it has lost something that it cannot or may not wish to live without. Despair is also viewed as being analogous to states of hopelessness, helplessness, and depression (Harlow et al., 1963; Kaufman & Rosenblum,

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1967; Rutter, 1972). However, it appears that attachment behavior on the part of the infant, and indeed any underlying construct used to account for the behavior (bonds), may not be critical factors for understanding how the infant responds to separation.

Infants attach to objects regardless of their survival value. A particular set of sensory characteristics associated with contact comfort is the critical releaser of attachment behavior in rhesus infants (Harlow, 1958), and surrogate-reared infants exhibit attachment behavior that is comparable to that of mother-reared infants. Nevertheless, a terry cloth covered pedestal has little value to the infant as defined by any of the parameters pertinent to the relationship of mother-reared infants to their mothers. The infants are not physiologically dependent on their surrogates. All of their nutritional and hygienic requirements are provided by human caretakers. The surrogate also does not comfort the infant and it does not protect it. The infant is left to fend for itself in these respects. Yet in the present study the surrogate-reared infants had primarily a despair-like response to separation. Therefore, despair following separation does not appear to be a consequence associated with a relatively greater objective value of the attachment object, positive nature of the mother-infant relationship, or loss of security following separation.

Consideration of the circumstances and results in the present study suggests that over and above serving as an attachment object, a biological mother alters the neurobiological characteristics of the infant, and this may account for the way in which the infant confronts the separation environment. A second and related possibility is that the neurobiological characteristics of the infant may be affected by the nature of the environment that exists for the infant if a mother is not present, or if the mother does not protect the infant. Indeed, a conclusion that can be gleaned from a series of studies by Hinde and colleagues is that mother-reared infants are most disturbed by separation when the circumstances surrounding the separation degrade the reliability and adequacy of the mother (Hinde, 1977; McGinnis, 1978).

Psychobiological Attachment

In a series of studies spanning several decades, Hofer has shown that the biological mother has a significant effect on the neurobiological characteristics of the mammalian infant (Hofer, 1987). In rhesus monkeys these effects are manifested in part by increased CSF NE before and after separation. Mother rearing is also associated with characteristic neurobiological responses to social environmental transitions that persist well into the second year of life (Kraemer et al., 1989). Thus, there are neurobiological characteristics of the infant that are attributable to rearing with a biological mother, and they persist in the mother's absence.

The mother initially provides an external regulatory mechanism for many of the physiological mechanisms that the infant possesses but does not regulate itself. These effects are mediated by effects of the mother on the infant's neurobiological processes (Hofer, 1987; Reite, 1987; Reite & Capatanio, 1985). At some point in development the infant becomes self-regulating through the development of internal regulatory mechanisms entrained to the stimuli that the mother provides. Field (1985) characterized this process, involving both behavioral and neurobiological changes, as *psychobiological attunement*. Once the infant is tuned to the mother its neurobiological characteristics are different than they were before.

The internalized regulatory characteristics of the infant that develop in relation to the mother will be referred to as a "Mother Icon" in order to distinguish the concept from Bowlby's working model of the mother. As a construct, the Mother Icon is an internalized neurobiological guidance and regulatory system that develops in relation to the adult mother. As an entity, the Mother Icon is a pattern of "experience-expectant" neurobiological development that is determined by the behavioral characteristics of the mother (Greenough, Black, & Wallace, 1987). Once the infant has become self-regulating in relation to the mother, it cannot be separated from critical regulatory aspects of the mother. It takes the Mother Icon with it to the new environment.

This analysis suggests that interactions with a mother or caretaker determine the regulatory characteristics of the infant, and it is the behavioral/neurobiological regulatory capacity of the infant that determines the nature of the response to separation. If so, there must be critical differences between mother- and surrogate-reared infants even while the infants are with their attachment objects and exhibiting comparable levels of attachment behavior. In this regard it is

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noted that surrogate-reared infants have low levels of CSF NE prior to separation. One possibility is that surrogate as opposed to mother rearing allows development of the NE system to fail in some way. For instance, the mother plays an initial role in nutritional and thermoregulation of the infant, and removal of these regulatory aspects of the mother is associated with variation in the response to separation and changes in brain biogenic amine systems in rat pups (Stone, Bonnet, & Hofer, 1976). Another possibility is that there are features of the preseparation environment that could alter NE system activity.

Surrogate Rearing

The results of other studies of surrogate-reared infants are consistent with the finding that the attachment behavior of surrogate-reared infants is comparable to that of mother-reared infants (Mason & Berkson, 1975; Mason & Capitanio, 1988; Meyer et al., 1975; Reite, Short, & Seiler, 1978). However, upon separation, mother-reared infants have increased plasma cortisol and often exhibit changes in sleep cycles and thermoregulation, whereas these effects have not been observed in surrogate-reared infants (Meyer et al., 1975; Reite et al., 1978).

Some of these findings are to be expected because surrogate-infant separation does not involve a removal of any physiological regulatory characteristics of a mother. However, these findings have also been interpreted as showing that, despite comparable attachment behavior, surrogate-reared infants are not as attached to their surrogates as mother-reared infants are attached to their mothers. Another possibility is that surrogate-reared infants respond or fail to respond to a quite different set of environmental contingencies, and this may be observable at both behavioral and biological levels of analysis.

Protest is most profitably thought of as searching or signaling behavior that is likely to increase the probability of mother-infant reunion (Thierry, Steru, Chermat, & Simon, 1984). Its exhibition appears to depend on experience with behavior-contingent responses on the part of the mother (Mason & Capitanio, 1988). If a surrogate never responds to the infant's behavior, then the value of protest behavior is minimal, nor would the infant have any experience prior to separation that would reinforce it. In the present study, even modest increases in responsiveness provided by a rocking surrogate were associated with increases in protest-like behavior.

With regard to despair-like behavior, many studies have employed at least one condition in which the infant is separated from the surrogate and placed in a different environment, but there are few reports of despair-like behavior. However, Harlow (1958) provided a picture of the response to surrogate-infant separation that is remarkably similar to that observed in the present study. It is also true that this response was ameliorated if the surrogate was present in the new environment, so there are significant differences in the way in which surrogate-reared infants are reported to respond to separation. A commonality across studies is that surrogate-reared infants respond to separation differently than mother-reared infants studied at the same time. This leads to the speculation that variation in the laboratory environment and experimental protocol may be critical factors for surrogate- as opposed to mother-reared infants.

Security

Surrogate-reared infants must deal with human caretakers, and they also have visual and auditory contact with other monkeys. Rhesus infants over 3 months of age commonly treat the former, and often the latter, as threats (Kalin & Shelton, 1989). Confrontations range from simply having observers in the room, to husbandry procedures such as cage cleaning, swapping of terry cloth covers on the surrogate, treatment for disease, to experimental procedures. Most experiments are designed to insure that the infant cannot predict when a relatively innocuous or another somewhat traumatic event will occur. Environmental transition or separation may increase uncertainty even though the environment has not changed in other respects.

This suggests that the surrogate-reared infant can be repeatedly exposed to a variety of unpredictable, uncontrollable, and unavoidable stressors that quantitatively if not qualitatively exceed those to which most mother-reared infants are exposed. While it is not possible to estimate how this exposure actually varies across studies, there is a neurobiological correlate of exposure to such stressors. Repeated exposure to unpredictable and uncontrollable stressors is associated with depletion of brain NE (Stone, 1983; Weiss & Simson, 1986), and the surrogate-reared infants in this study had

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low CSF NE prior to separation. Furthermore, the sensitivity of infants reared without mothers to drugs that deplete NE, and the ability of antidepressant drugs that reduce NE reuptake to block despair, parallels similar effects in rodents and monkeys known to have depletions of brain NE (Hrdina, Von Kulmiz, & Stretch, 1979; Kraemer & McKinney, 1979; Kraemer et al., 1981, 1984a, 1984b; Suomi, Seaman, Lewis, DeLizio, & McKinney, 1978). Therefore, the data from various studies are compatible with associating despair with behavioral and neurobiological aspects of repeated exposure to stressors. However, reduced NE system activity and an inability to mobilize active behavioral responses to environmental stressors may exist prior to separation.

In this regard, it is noted that increasing the responsiveness of the surrogate, rocking versus stationary, does alter the response to separation toward being more protest-like, but this is not associated with increases in *baseline* CSF NE. This may indicate that responsiveness of the "mother" is important, but responsiveness in itself is not pertinent to another role of the mother, which is protection of the infant from environmental stressors. This is pertinent to reconsideration of the meaning of security and safety as far as the infant is concerned.

The fact that infant rhesus monkeys can develop an affiliative relationship with an inanimate object shows that attachment behavior is not tuned to the survival value or secure base characteristics of the attachment object. It is also quite likely that "attachment schemata" vary across species and, by comparison to nonhuman primates, humans may progressively attend to visual and auditory stimuli over and above somatosensory input. Human infants will not accept an inanimate object as a mother substitute (Rutter, 1979a). Yet the same attachment/safety dissociation may prevail, with the infant attending to characteristics of the mother that have no causal linkage to how safe the infant will be with that mother. This may work most of the time because it is unlikely that the characteristics or quality of maternal care and risk for exposure to stressors are independent in almost any natural setting. The cluster of maternal characteristics associated with secure attachment also usually protects the infant from traumatic experiences. Secure attachment and safety of the infant seem to go hand in hand. Conversely, the cluster of maternal characteristics associated with insecure or anxious attachment probably insures that there has been or will be an increased exposure to events that are traumatic for the infant (Bowlby, 1980, pp. 311–319; Rutter, 1979a). In this light, the caretaking of children in an institutional setting should be considered.

While the circumstances surrounding rearing of rhesus infants in a laboratory bear little relation to rearing conditions in a feral environment, they seem to be quite pertinent to those encountered by institutionalized children and by children of indifferent or abusive mothers. In this case, the results suggest that these children may be vulnerable to deflections in behavioral/neurobiological development not only because they may not have a compassionate and competent caretaker, but because they may endure increased exposure to environmental stressors that take on added significance if they must be endured in the absence of caretaker who also protects the infant.

Conclusion and Summary

Despair does not appear to be simply a behavioral response to separation, although it is most conspicuous in this context. In laboratory-reared rhesus infants, the major determinant of risk for despair is environmental and depends on the characteristics of the attachment object. The effects of this environmental factor appear to be translated into differences in neurobiological status over the course of early development. Risk for despair is associated with low CSF NE, but this study provides no evidence that this is related to an underlying and independent neurobiological risk factor in otherwise healthy rhesus infants.

Attachment behavior does not appear to alter brain NE system activity, or the nature of the response to separation. Instead, the results of this study add to the increasing body of evidence that maternal contact alters the neurobiological characteristics of the infant. The neurobiological changes attributable to regulatory effects of the mother, collectively referred to as the Mother Icon, may play a significant role in determining how the infant deals with the separation environment. To the degree that the mother is a competent and survivable member of the species, the infant may share these characteristics in relation to its genetic endowment. However, if the mother does not protect the infant from environmental stressors, this may also contribute to degradation of behavioral and biological capacity to deal with separation.

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The component of variation in response to separation that cannot be attributed to environmental factors appears to be expressed in lability or stability of neurobiological and associated behavioral characteristics of the infant. This finding may have implications with regard to the concept of individual differences in temperament, that is, how well the infant can become tuned to its caretaker, and/or how reliable and persistent the effects of this process will be.

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